

Question

The following chemical processes are described:

1. In an aqueous solution of ethylenediamine, metallic potassium reacts with KReO_4 to yield a hydride containing Re 68%.
2. Ceric ammonium nitrate can be used to determine the content of sodium nitrite.
3. Aluminum carbide can react with a hydrochloric acid solution of mercury(II) chloride to produce a highly toxic substance.
4. CO , B_2H_6 react in a 2:1 ratio in tetrahydrofuran to form a highly symmetrical cyclic molecule.
5. Chloroplatinic acid can accelerate the hydrolysis reaction of sodium borohydride.
6. Cesium azide can react with N_5SbF_6 in sulfur dioxide solvent, with the expectation of obtaining an N_8 molecule, but only a common gas is produced.

Regarding the main chemical or ionic equations (with stoichiometric coefficients in their simplest integer ratios) involved in these six processes, which of the following statements is correct?

- A.** In the chemical equation for Process 1, the sum of the coefficients of the reactants is less than 30.
- B.** In the chemical equation of process 2, the sum of the coefficients of the products is 10.
- C.** In the chemical equation of process 3, the sum of the coefficients of the reactants is less than 20
- D.** In process 4, a certain product molecule generated has 12 atoms.
- E.** In the ionic equation of Process 5 (the products of this ionic equation need to reflect the coordination state of the elements), the sum of the coefficients of the products is 19.

F. In the chemical equation of process 6, the sum of the coefficients of the products is an even number.

G. None of the above options are correct

Answer

Correct Answer: **E**

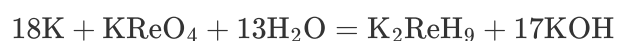
Detailed Explanation

Process 1: Based on the mass fraction, it is easy to deduce that the hydride is K_2ReH_9 ; it is observed that the oxidation state of Re does not change; hydrogen in the hydride has a -1 oxidation state, so potassium acts as the reducing agent; thus, the redox equation is balanced:

CHECKPOINT

1 PTS

The hydride is K_2ReH_9



CHECKPOINT

1 PTS



The sum of the reactant coefficients is 32, so option A is incorrect.

Process 2: Ceric ammonium nitrate acts as the oxidizing agent, oxidizing nitrite to nitrate while being reduced to cerium(III) nitrate. The balanced chemical equation is:



CHECKPOINT

1 PTS



The sum of the product coefficients is 9, so option B is incorrect.

Process 3: Aluminum carbide reacts with hydrochloric acid, where the negatively charged carbon bonds with the positively charged hydrogen; the highly toxic product formed should contain Hg(II), thus likely being dimethylmercury. The balanced equation is:

CHECKPOINT

1 PTS

The highly toxic product is dimethylmercury $\text{Hg}(\text{CH}_3)_2$



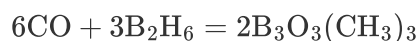
CHECKPOINT

1 PTS



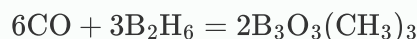
The sum of the reactant coefficients is 22, so option C is incorrect.

Process 4: This is a classic reaction for synthesizing boroxine analogs:



CHECKPOINT

1 PTS



The product contains 18 atoms, so option D is incorrect.

Process 5: Chloroplatinic acid accelerates the hydrolysis of sodium borohydride, which produces borate ion $\text{B}(\text{OH})_4^-$ and hydrogen gas. With the addition of chloroplatinic acid, hydride ions reduce platinum(IV) to platinum metal. The balanced redox equation is:

CHECKPOINT

1 PTS

Sodium borohydride hydrolysis produces borate ion $\text{B}(\text{OH})_4^-$, and hydride ions reduce platinum(IV) to platinum metal



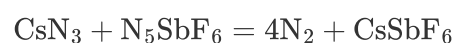
CHECKPOINT

1 PTS



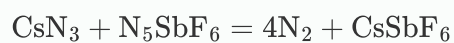
The sum of the product coefficients is 19, so option E is correct.

Process 6: The reaction was expected to produce N_8 from the reaction of azide ion with N_5^+ , but the actual common gas produced is nitrogen. Thus, the equation is:



CHECKPOINT

1 PTS



The sum of the product coefficients is 5, an odd number, so option F is incorrect.

In conclusion, option E is correct.